

Technology-Enhanced Strategies to Address Learning Needs in Undergraduate Music Theory: A Systematic Review

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Abstract

In response to the increasing emphasis on student-centered learning in higher education, this systematic literature review explores the evolving learning needs of students enrolled in undergraduate music theory courses. Drawing on empirical studies published between 2015 and 2024 from Scopus, Web of Science, Eric, JSTOR, and IEEE databases, the review synthesizes current evidence on how music theory instruction addresses cognitive, affective, and contextual student needs. A total of 15 high-quality peer-reviewed articles were selected based on predefined inclusion criteria and assessed using the Critical Appraisal Skills Programme (CASP). The analysis is organized around four key dimensions: the integration of educational technology, instructional strategies, pedagogical challenges, and assessment practices. Findings highlight a growing trend toward the use of mobile applications, flipped classrooms, and differentiated instruction; all aimed at enhancing learner autonomy, engagement, and conceptual understanding. However, key barriers remain, including limited resources, abstract content, and unequal technological access. The study concludes with evidence-informed recommendations for educators, curriculum designers, and policymakers to promote inclusive, adaptive, and pedagogically coherent models of music theory education.

Keywords

technology-enhanced learning, music theory education, student learning needs, personalized instruction, digital pedagogy

Introduction

In the era of increasingly globalized, technology-mediated, and learner-diverse educational landscapes, personalized instruction and student-centered learning have become foundational principles in curriculum design and pedagogical reform (Whalley et al., 2021). These paradigms emphasize the importance of accommodating individual learning preferences, cognitive styles, and socio-emotional motivations within instructional systems that are both inclusive and adaptive. Within this evolving context, music education plays a pivotal role—not only in fostering artistic and emotional development but also in cultivating critical thinking, creativity, and cultural literacy (Cui, 2023). Yet, traditional music education has been predominantly anchored in performance-oriented and skills-based pedagogies, which often prioritize uniformity over learner variability.

As higher education seeks to respond to student diversity, an increasing number of studies point to a

pedagogical misalignment between conventional approaches and the distinctive demands of music theory instruction. Unlike performance-based learning, which provides immediate sensory feedback and embodied engagement, music theory is inherently abstract, symbolic, and analytically complex, encompassing domains such as harmonic function, formal structure, and symbolic notation (Sternberg & Kibelsbeck, 2022). These characteristics create substantial learning barriers when instruction is delivered in standardized formats lacking

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Data Availability Statement included at the end of the article



differentiation (Gheysens et al., 2022). Learners' diverse pacing, metacognitive strategies, and prior knowledge further highlight the need for responsive pedagogies capable of supporting multiple modes of engagement and comprehension.

Although differentiated instruction is increasingly recognized as a cornerstone of effective music education, empirical evidence focusing specifically on the learning needs of students in music theory courses remains limited, fragmented, and methodologically inconsistent. Much of the existing scholarship emphasizes performance training or relies on anecdotal reports and localized case studies, restricting the generalizability of findings for broader curriculum reform (Quaglia, 2015; Romiszowski, 2016). Furthermore, despite the expansion of digital technologies and online platforms that enable personalized pathways, few studies have systematically examined how these tools are employed in music theory instruction to address students' cognitive and affective needs. This gap is notable given that music theory is both central to undergraduate curricula and particularly vulnerable to disengagement when pedagogical approaches fail to account for learner diversity.

To address this gap, the present study conducts a systematic review of empirical literature published between 2015 and 2024, with a specific focus on undergraduate music theory education. Drawing on peer-reviewed studies indexed in Scopus, Web of Science, IEEE, Eric and JSTOR, the review synthesizes evidence in four domains: (1) The application of technology in music theory education, (2) Teaching methods and strategies in music education, (3) Challenges and solutions in music theory education, and (4) Assessment and effectiveness of music theory education. By doing so, the study provides an evidence-informed foundation for developing inclusive, student-responsive, and theoretically grounded music curricula. It contributes to academic discourse on differentiated music pedagogy while offering practical insights for educators, curriculum developers, and policymakers seeking to improve learning outcomes in music theory education.

Theoretical Background

Music education is widely acknowledged as a key medium for fostering students' esthetic awareness, critical thinking, and creative potential (Guo et al., 2020). However, contemporary educational contexts—characterized by increased student diversity and rapid technological change—present new challenges for traditional instructional models in music education (Zhang, 2024). To respond effectively, music curricula must promote learner autonomy, participatory learning, and individualized support (Cox, 2007). A robust understanding

of students' learning needs is thus essential for enhancing instructional quality and improving learning outcomes (Johnson et al., 2023).

Learning needs are broadly defined in educational research as students' expectations and requirements regarding instructional content, methods, and resources necessary to engage effectively and achieve success in their learning process (Roy et al., 2013). In the context of music theory, these needs are primarily centered around cognitively demanding tasks such as symbolic notation, harmonic analysis, and the abstraction of compositional structure (Sternberg & Kibelsbeck, 2022). These challenges often arise not from a lack of interest, but from the abstract nature of the subject, which requires sophisticated cognitive processes and mental models that may not align with students' previous knowledge. Addressing these learning needs demands teaching methods that are differentiated, responsive, and capable of bridging cognitive gaps. Additionally, the integration of digital tools and online learning environments has both enhanced opportunities for personalized learning and introduced complexities that require careful pedagogical adaptation (Crawford, 2017).

Music education typically includes two interrelated dimensions: theoretical instruction (e.g., music history, harmonic functions, and compositional frameworks) and practical training (e.g., instrumental performance, vocal technique, and improvisation; Pellegrino et al., 2019; Qisen et al., 2024). Students' preferences and learning profiles often vary significantly across these domains, highlighting the need for flexible instructional design that accommodates diverse interests and proficiencies (Anderson, 2022). Systematic inquiry into student feedback—through surveys, interviews, and assessments—can provide crucial insights into these needs and inform evidence-based pedagogical adjustments (Jääskeläinen et al., 2023; Salvador & Culp, 2022). Furthermore, systematic evaluation methodologies allow researchers to aggregate empirical evidence, identify consistent patterns, and minimize the limitations of individual research designs, thereby offering a broader understanding of how instructional strategies, technological tools, and institutional factors influence students' academic experiences (Pollock & Berge, 2018; Siddaway et al., 2019). This review applies these systematic techniques to explore the multifaceted nature of learning needs in music theory education and to guide pedagogical innovation in this field.

Purpose of the Study

The primary aim of this study is to conduct a comprehensive systematic literature review of empirical research addressing student learning needs in undergraduate

music theory education. The review investigates how these needs have been defined, identified, and addressed across varying educational contexts and instructional models. By synthesizing results from selected peer-reviewed studies published between 2015 and 2024, the research seeks to map the scope of learning difficulties, individual preferences, and contextual challenges reported by students and instructors.

A particular focus is placed on the pedagogical approaches employed in response to these needs, including the use of technology-enhanced learning, flipped classrooms, differentiated instruction, and student-centered frameworks. The study further explores how digital platforms and flexible learning environments impact student engagement with theoretical content. Rather than prescribing universal solutions, this review provides a structured synthesis of the current knowledge base, highlighting dominant themes, recurring gaps, and potential directions for future research and curricular development in music theory education.

Methodology

Research Question

This systematic review aims to address the following question: “What do empirical studies published between 2015 and 2024 reveal about the learning needs of students in music theory courses?”

To answer this central question, the following research questions (RQs) were formulated:

RQ1: How is technology applied in music theory education?

RQ2: What teaching methods and strategies are used in music theory education?

RQ3: What challenges do students encounter, and what solutions are implemented in music theory education?

RQ4: How is student learning assessed, and how effective is music theory education?

Search Strategy

In the context of systematic literature reviews, the PICO framework—originally developed for use in health and clinical research—serves as a valuable tool for structuring and refining research questions. Although traditionally associated with evidence-based medicine, its logical structure has been increasingly adapted for use in education and the social sciences. PICO helps delineate the core elements of a research inquiry, thereby enhancing clarity, focus, and methodological rigor Table 1 (Methley et al., 2014).

This study adopts a systematic review approach to search and screen relevant literature on the investigation of students’ learning needs in music courses. Articles were searched in databases such as Scopus, Web of Science (WoS), Eric, JSTOR, and IEEE, using the following keywords: (“music theory course*” OR “music theory education” OR “music theory”) AND (“learning need*” OR “learning preference*” OR “educational need*” OR “student need*” OR “requirement*”). These keywords were then entered to search for specific studies within the Scopus, WoS, Eric, JSTOR, and IEEE databases. Although the primary focus of this review is undergraduate music theory education, studies involving other participant groups (e.g., music teachers or adult musicians) were also considered when they provided relevant pedagogical insights into the teaching and learning of music theory concepts. Such studies were retained because they illuminate instructional practices, learning strategies, and knowledge transmission processes that are transferable to undergraduate music theory contexts. The search period was set to cover publications from 2015 to 2024. As presented in Table 2, a total of 201 articles were eventually identified. All retrieved results were imported into Zotero for automatic deduplication, followed by independent manual verification by two researchers to ensure accuracy. Title and abstract screening were independently conducted by two researchers, with a calibration exercise performed on 30 randomly selected studies before formal screening to unify inclusion and exclusion criteria. During the formal screening process, inter-rater reliability was assessed using Cohen’s

Table 1. Question Formulated Based on the PICO Methodology.

P = Population	I = Intervention	C = Comparison intervention	O = Outcome
Undergraduate students in music theory courses	Instructional strategies, educational technologies, or curriculum models	Traditional or undifferentiated instruction	Learning needs, academic engagement, instructional effectiveness

Source. Schiavenato and Chu (2021).

Table 2. Web of Science, Scopus, and IEEE Keyword.

Journal search	Search key strings content
Web of Science	TS= ("music theory* course* OR "music theory* education" OR "music theory*") AND TS= ("learn* need*" OR "learn* preference*" OR "educational need*" OR "student need*" OR "need*" OR "requirement*")
Scopus	TITLE-ABS-KEY ("music theory* course*" OR "music theory* education" OR "music theory*") AND TITLE-ABS-KEY ("learn* need*" OR "learn* preference*" OR "educational need*" OR "student need*" OR "need*" OR "requirement*") AND PUBYEAR > 2015 AND PUBYEAR < 2025
IEEE	("music theory course*" OR "music theory education" OR "music theory") AND ("learning need*" OR "learning preference*" OR "educational need*" OR "student need*" OR "requirement*")
Eric	("music theory course*" OR "music theory education" OR "music theory") AND ("learning need*" OR "learning preference*" OR "educational need*" OR "student need*" OR "requirement*")
JSTOR	("music theory course*" OR "music theory education" OR "music theory") AND ("learning need*" OR "learning preference*" OR "educational need*" OR "student need*" OR "requirement*")

Table 3. Inclusion and Exclusion Criteria for Publications.

Inclusion criteria	Exclusion criteria
Released 2015–2024	Released before 2015
Published in Wos, Scopus, Eric, JSTOR and IEEE	Published in databases other than Wos, Scopus, Eric, JSTOR, and IEEE
Published in peer-reviewed journals	Not published in a peer-reviewed journal
Downloadable documents	Unable to download the literature
Regular literature journals	Gray literature (e.g., theses, dissertations, books, government reports)
Is not a study of a survey of student learning needs in music programs	Not a study on the survey of students' learning needs in music courses

κ coefficient (target $\kappa \geq 0.70$). If $\kappa < 0.70$, the criteria were revised after further discussion. Any disagreements were resolved through arbitration by a third researcher.

This paper aims to provide research on the investigation of students' learning needs in music courses through a systematic literature review and seeks to deepen the understanding of relevant issues in this field. The study follows the methodology of a systematic review. As shown in Table 3, the review process includes identification, screening, eligibility, inclusion, and exclusion.

Assessment of Study Quality

This study employed the Critical Appraisal Skills Programme (CASP) tool to assess the quality of included literature. CASP is a systematic evaluation tool designed to assist researchers in critically appraising the quality of various studies across design, methodology, analysis, and reporting aspects, thereby determining the reliability, relevance, and validity of their findings (Long et al., 2020).

Following this rigorous evaluation procedure, a total of 15 studies were deemed to meet the predefined quality thresholds and were subsequently included in the final synthesis, confirming their compliance with CASP criteria in terms of methodological robustness, validity, and overall research integrity. The detailed appraisal

outcomes are presented in Table 4. To enhance clarity and facilitate interpretation, a color-coded scheme was adopted in the table: green indicates that a study clearly satisfied the corresponding CASP criterion ("Yes"), yellow denotes cases where the available information was insufficient or ambiguous ("Can't tell"), and red represents criteria that were not met ("No"). This visual approach enables transparent comparison of methodological quality across studies and supports a more nuanced understanding of strengths and limitations within the included literature.

A comprehensive description of the appraisal procedure and detailed scoring justifications for each study are provided in the Supplemental Appendix.

PRISMA Flowchart for Screen of Studies

This systematic literature review was conducted in accordance with the PRISMA 2020 (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines, to ensure methodological transparency, analytical rigor, and procedural replicability. The aim of the review was to synthesize empirical studies that examine students' learning needs in undergraduate music theory education, particularly within the context of pedagogical innovation, differentiated instruction, and technological integration in higher education. To achieve this

Table 4. Quality Assessment Results of Each Included Article Using the CASP Tool.

CASP Assessment Criteria	Nedelcuț and Ghișa (2020)	Buonviri and Paney (2015)	Barić and Begić (2022)	Kavčić Pucihar et al. (2024)	Duker et al. (2015)	Montoya Rubio (2021)	Cayari (2022)	Li (2020)	Pesek et al. (2022)	Buonviri and Paney (2020)	Uludag and Satir (2025)	Nolte (2019)	Gutierrez (2018)	Attas (2018)
Valid aim?	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes
Recruitment acceptable?	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes
Exposure correctly measured?	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes
Outcome correctly measured?	yes	yes	canit tell	yes	yes	yes	yes	yes	yes	yes	yes	canit tell	yes	yes
Identified confounding?	yes	canit tell	canit tell	yes	canit tell	canit tell	canit tell	canit tell	yes	canit tell	yes	canit tell	yes	yes
Addressed confounding?	yes	canit tell	canit tell	yes	yes	canit tell	canit tell	yes	yes	yes	yes	yes	no	yes
Subject follow-ups finish enough?	canit tell	canit tell	canit tell	canit tell	canit tell	canit tell	canit tell	yes	yes	yes	yes	no	canit tell	canit tell
Subject follow-up long enough?	canit tell	canit tell	canit tell	canit tell	canit tell	canit tell	canit tell	canit tell	canit tell	yes	canit tell	yes	yes	yes
Results believable?	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	canit tell	canit tell
Results comparable to others?	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes
Implication for practice?	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes

objective, a multi-phase and iterative screening process was implemented, combining both automated and manual techniques for identifying, filtering, and selecting relevant studies.

To ensure methodological rigor and provide comprehensive coverage of research at the intersection of education, music pedagogy, and instructional technology, this study utilized five interdisciplinary databases: Web of Science, Scopus, Eric, JSTOR, and IEEE. These databases were prioritized due to their inclusion of high-impact, peer-reviewed journals from social sciences and applied technology fields, ensuring reliable and diverse literature coverage. The search period was confined to publications between January 2015 and December 2024, capturing contemporary trends such as the integration of digital platforms, the shift toward student-centered learning, and the impacts of the COVID-19 pandemic on remote and blended education.

The initial search identified a total of 502 records: Scopus ($n = 111$), Web of Science ($n = 82$), IEEE ($n = 8$), Eric ($n = 13$), and JSTOR ($n = 288$). Upon importing the records into the Zotero reference manager, 121 duplicates were automatically flagged by the system. After manual verification by two independent reviewers, 381 unique articles records remained. The screening process followed the PRISMA 2020 guidelines, which were executed in three phases. In the first phase, title screening excluded 236 irrelevant records, leaving 145 records to be reviewed based on their abstract. In the second phase, abstract screening, 86 records were excluded for not addressing music pedagogy or instructional technology, leaving 59 records to proceed to full-text evaluation. In the final phase, full-text screening, 44 records were excluded. The reasons for exclusion encompassed: (a) articles that did not belong to the “Education” research category; (b) those lacking keywords related to music teaching and learning; (c) those not focusing on technology integration or innovative teaching methods in music theory; (d) non-peer-reviewed publications or gray literature (e.g., conference proceedings, theses, dissertations); and (e) articles that were inaccessible for full review.

Following these exclusions, 15 full-text articles were retained for final analysis. Discrepancies between the reviewers were resolved through discussion and consensus. Ultimately, 15 studies met all inclusion criteria and were incorporated into the final synthesis. A detailed visualization of the entire process is provided in Figure 1 (PRISMA flow diagram).

Data Extraction

For each of the Fifteen included studies, a structured data extraction protocol was applied to systematically record key characteristics. The extracted variables

included author(s), year of publication, research aims, participant demographics, methodological design, instructional interventions, and findings related to learning needs in music theory education. A combined approach of inductive and deductive thematic analysis was adopted. Inductive coding allowed patterns and themes to emerge directly from the data, while deductive coding ensured alignment with relevant theoretical frameworks, specifically Universal Design for Learning (UDL) and Self-Determination Theory (SDT). Coding was conducted manually using Microsoft Excel. To enhance the reliability of the analysis, two researchers independently coded the materials and subsequently conducted a cross-review of each other’s work. Discrepancies in coding interpretation were discussed and resolved by consensus.

This rigorous, multi-phase procedure ensured that all included studies met high standards of empirical quality and thematic relevance. The resulting synthesis offers a robust and theory-informed evidence base for understanding the nature, scope, and instructional implications of students’ learning needs in contemporary music theory education.

The final selection of 15 studies (2015–2024) resulted from rigorous quality screening and domain-specific constraints. Music theory’s abstract nature—evidenced by harmonic analysis and symbolic notation challenges—reduces empirical outputs versus performance-based subjects (Sternberg & Kibelsbeck, 2022), with only 5.7% of music education publications focusing on theory pedagogy (Yang & Welch, 2023). Table 5 summarizes these high-evidence studies representing current best practices.

Data Synthesis

Given the methodological diversity of the included studies, a statistical meta-analysis was not undertaken. The selected articles employed a range of research designs (qualitative, quantitative, and mixed methods), as well as varied intervention types and outcome measures, which limited the feasibility of pooling effect sizes within a single quantitative model. Accordingly, this review adopted a narrative and thematic synthesis approach, which is widely recommended for systematic reviews involving heterogeneous educational evidence (Siddaway et al., 2019).

The synthesis proceeded in two stages. First, a descriptive mapping was conducted to summarize the key characteristics of the included studies, including research design, participant profiles, and instructional contexts. Second, a thematic synthesis was undertaken to identify recurring patterns related to students’ learning needs in music theory education. The interpretation of themes was informed by relevant theoretical perspectives,

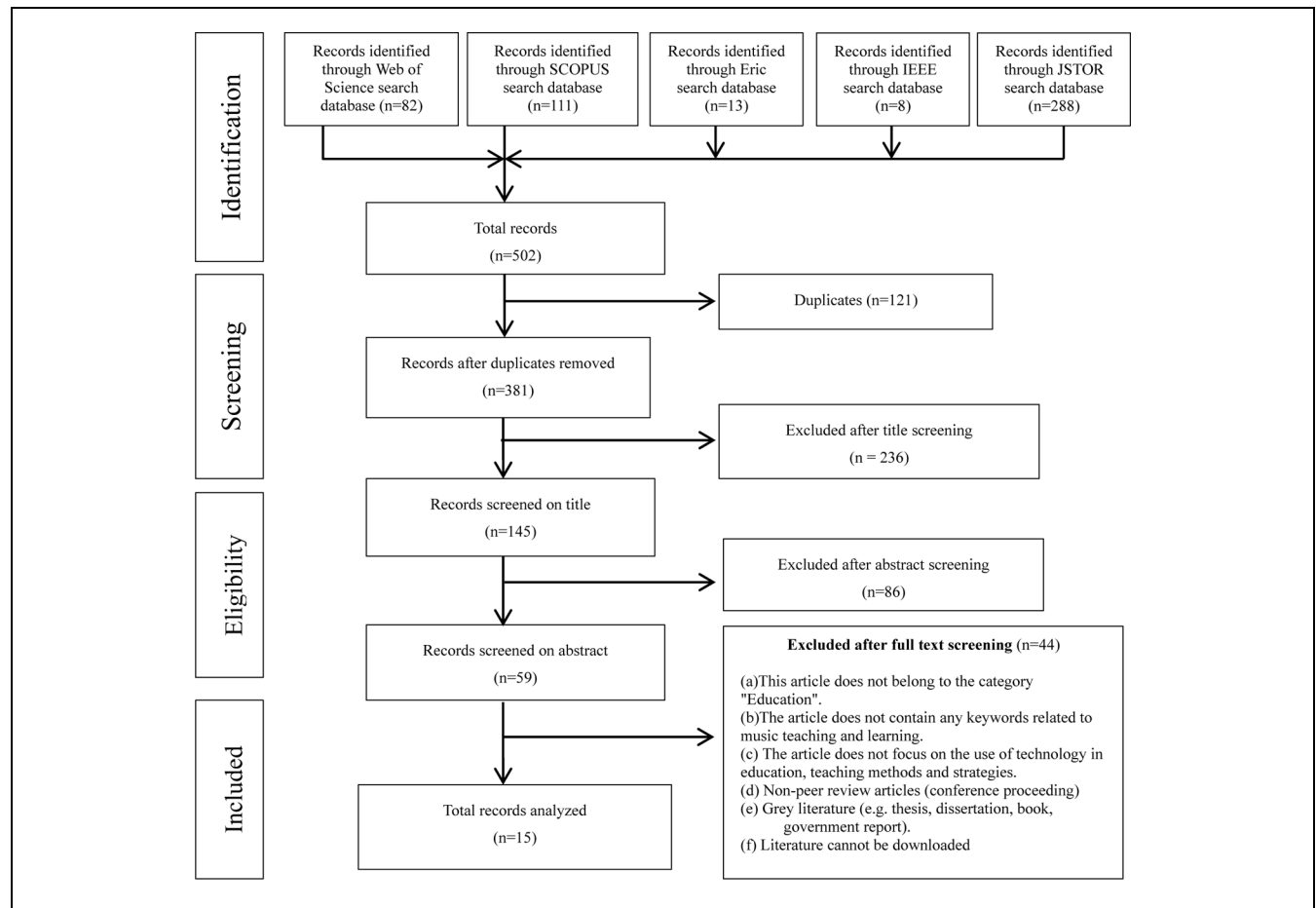


Figure 1. PRISMA diagram for the systematic review.

particularly Self-Determination Theory (SDT) and Universal Design for Learning (UDL), enabling the

integration of findings from diverse empirical studies while maintaining methodological transparency and analytical rigor.

Table 5. Selected Articles for the Systematic Analysis (n = 15).

No.	Author(-s)
1	Uludag and Satir (2025)
2	Buonviri and Paney (2020)
3	Pesek et al. (2022)
4	Li (2020)
5	Cayari (2022)
6	Montoya Rubio (2021)
7	Duker et al. (2015)
8	Kavčič Pucihar et al. (2024)
9	Ebbini (2023)
10	Barić and Begić (2022)
11	Buonviri and Paney (2015)
12	Nedelcuț and Ghișa (2020)
13	Nolte (2019)
14	Gutierrez (2018)
15	Attas (2018)

Source. Developed by the author.

Results

Although initial search results revealed 502 relevant studies, many of these were excluded after a rigorous screening and qualification review, as they did not align with the research topic of “investigating the learning needs in music theory courses.” Only research closely related to the educational needs of music students was ultimately included for analysis. The current analysis reveals that research on student learning needs in music theory courses is still an emerging field, with empirical studies being relatively limited. To the author’s knowledge, this area of research remains in its early stages. Among the 15 selected studies, six originated from the United States, two from Slovenia, and the remaining studies were from Turkey, China, Spain, Croatia, Jordan, Canada, and Romania. These studies employed diverse methodologies, including qualitative,

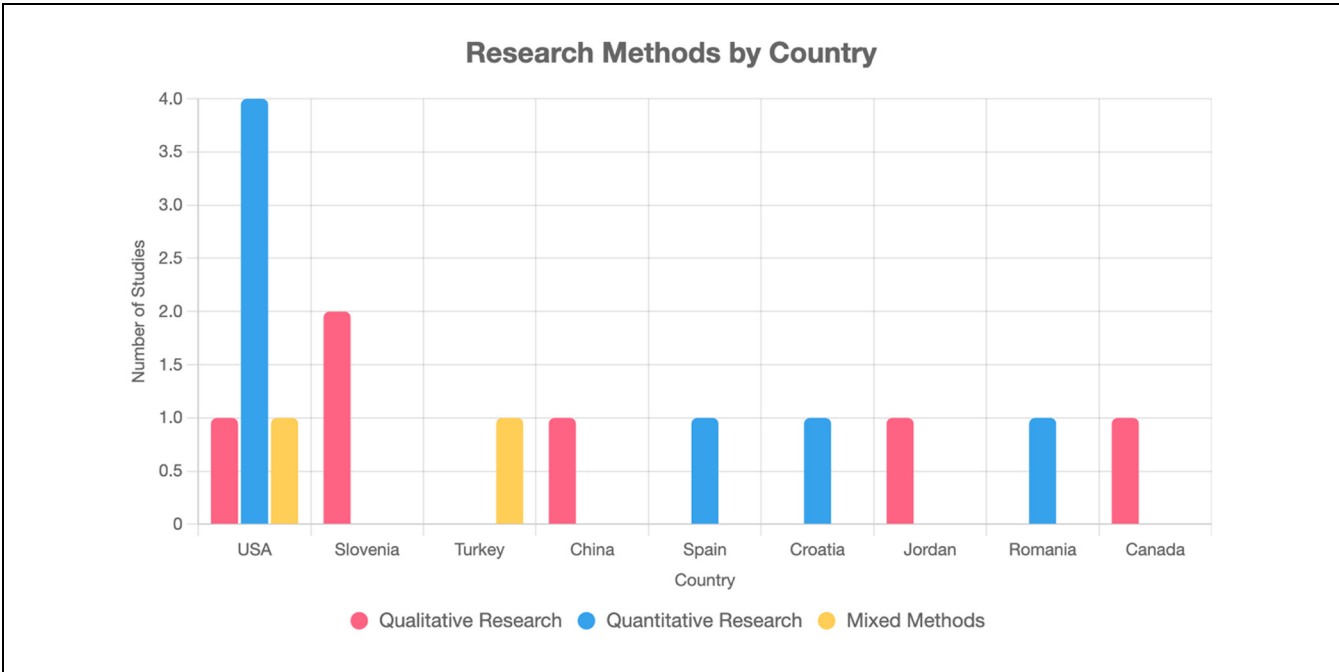


Figure 2. Distribution of methods and countries.

quantitative, and mixed methods approaches. Specifically, six studies utilized qualitative methods, seven employed quantitative methods, and two used a mixed-methods approach. The sample sizes ranged from 20 to 398 participants, including both students and music educators. Quantitative data were primarily obtained through questionnaires and experiments, while qualitative data were collected through interviews and case studies. Figure 2 illustrates the distribution of research methods and countries.

RQ1: How is Technology Applied in Music Theory Education?

Four studies focused on the use of technology in music theory education. A review and details of these four studies are shown in Table 6, including the year of publication, thematic focus, research objectives, geographic context, and details of methodological data for each article.

The reviewed studies highlight three primary technological applications in music theory education: (a) mobile applications, (b) auto-accompaniment software, and (c) online learning platforms. These tools enhance instruction by activating core learning mechanisms articulated in Self-Determination Theory (SDT) and Universal Design for Learning (UDL).

Mobile applications (e.g., Note Works, GarageBand) provide immediate feedback, interactive tasks, and gamified learning, which promote self-regulation and

strengthen autonomy and competence (Uludag & Satir, 2025). From a UDL perspective, their multimodal design offers multiple entry points to abstract concepts, improving accessibility. Auto-accompaniment software bridges theoretical knowledge with creative practice, enabling students to apply harmony and rhythm in authentic musical contexts, thereby reinforcing competence and stimulating expressive engagement (Li, 2020). Online platforms such as Troubadour extend individualized practice and sustain engagement, particularly in settings with limited traditional support (Pesek et al., 2022). However, their effectiveness depends on integration into coherent instructional design rather than standalone use.

Despite these benefits, adoption remains constrained by persistent barriers: inequitable access to technology, limited teacher training, and curricular rigidity (Buonviri & Paney, 2020). Future implementation should therefore prioritize pedagogical coherence, equitable access, and alignment with inclusive, culturally responsive teaching frameworks.

RQ2: What Teaching Methods and Strategies Are Used in Music Theory Education?

Three studies focused on pedagogical methods and strategies for music education. A review and details of the four studies are shown in Table 7, including year of publication, thematic focus, research objectives, geographical context, and details of methodological data for each article.

Table 6. Use of Technology in Music Theory Education.

Researcher/ Date	Country of origin	Methodology	Participants	Findings	Implications
Uludag and Satir (2025)	Turkish	Mixed methods study	83 students in the experimental group, 80 students in the control group	Mobile technology significantly improved music theory scores of students in the experimental group	Positive impact of mobile technology in music education
Buonviri and Paney (2020)	USA	Questionnaire	866 teachers (317 responded)	91% use digital technology to teach listening skills	Role and Challenges of Technology in Music Education
Pesek et al. (2022)	Slovenia	Experiments	20 students participated in the questionnaire and about 60 students used the platform	High level of student engagement during distance learning, reduced use during classroom learning	Value of the Troubadour platform as a complementary learning channel
Li (2020)	China	Experimental research	92 music majors	Computerized Automatic Accompaniment Improves Student Interest and Achievement	The Value of Computerized Automatic Accompaniment in Music Education

Table 7. Teaching Methods and Strategies in Music Education.

Researcher/ Date	Country of origin	Methodology	Participants	Findings	Implications
Cayari (2022)	USA	questionnaire	137 adult musicians who posted VGMs online	School-based music theory programs and informal music learning strategies are most helpful in posting VGMs	Music educators can expand their knowledge by studying the learning methods of musicians in the VGM community
Montoya Rubio (2021)	Spain	Questionnaire	409 music teachers in primary and secondary schools	Teachers highly endorse the use of Jaime Altozano's YouTube channel in the classroom	Need to generate pedagogical models to facilitate the integration of these videos in future teaching sequences
Duker et al. (2015)	USA	Literature Review and Case Study	Students in music theory classes	Teaching techniques such as standards-based grading, just-in-time instruction, and flipped classrooms can improve teaching effectiveness	These pedagogical techniques can be important tools in a variety of music programs.
Attas (2018)	Canada	Qualitative Study	Undergraduate students in the author's music theory courses	Explicitly teaching the expert process of analysis (e.g., through think-aloud lectures) improves students' analytical and writing abilities	Shifting pedagogy from product to process creates a more musical and effective learning environment

As digital technologies reshape music education, teaching strategies have gradually shifted toward more student-centered models. Although the primary focus of this review is undergraduate music theory education, several studies summarized in Table 7 involve participants beyond undergraduate students, including adult musicians engaged in online creative communities and teachers working in primary or secondary education. These

studies were retained because they provide empirically grounded insights into pedagogical practices and learning processes that are closely related to the development and transmission of music theory knowledge. Rather than examining undergraduate classrooms directly, they illuminate how theoretical understanding is acquired, practiced, and supported across different musical contexts. Consequently, their findings offer transferable

pedagogical implications that can inform instructional design in undergraduate music theory courses (Cayari, 2022; Montoya Rubio, 2021).

Within this broader evidence base, three pedagogical frameworks emerge as particularly influential for music theory instruction: standards-based grading (SBG), just-in-time teaching (JiTT), and the flipped classroom. Although these approaches originate from different educational settings, they converge on a common objective—supporting learners in mastering abstract and cognitively demanding theoretical concepts through more responsive and participatory instructional structures (Duker et al., 2015).

Standards-based grading (SBG) redefines assessment by aligning evaluation with clearly articulated learning outcomes rather than cumulative averages. In music theory, where conceptual progress can often appear opaque to learners, SBG decomposes complex analytical skills into discrete competencies, enabling students to identify both achieved and developing areas of understanding (Montoya Rubio, 2021). This transparency enhances learners' ability to monitor their own progress and reduces ambiguity surrounding assessment expectations. From a theoretical standpoint, SBG reinforces the competence dimension of Self-Determination Theory (SDT), as measurable progress strengthens students' confidence, while also aligning with Universal Design for Learning (UDL) by clarifying learning goals and evaluation criteria.

Just-in-time teaching (JiTT) further strengthens the alignment between preparation and instruction. Through short pre-class activities, instructors can identify conceptual difficulties and adapt in-class instruction accordingly (Cayari, 2022). In the context of music theory, this approach enables teachers to address challenging concepts—such as harmonic function, voice leading, or formal analysis—precisely when students encounter them. JiTT therefore enhances instructional responsiveness and supports differentiated engagement. In theoretical terms, the approach embodies UDL principles by offering multiple entry points to content, while also reinforcing SDT's principle of relatedness through continuous interaction between students and instructors.

The flipped classroom restructures the instructional sequence by relocating initial exposure to theoretical concepts—such as definitions, demonstrations, and analytical explanations—to out-of-class learning environments. This shift allows classroom time to focus on collaborative analysis, problem-solving, and applied musical practice (Duker et al., 2015). Such an arrangement is particularly effective for strategies that require guided analytical modeling, including think-aloud demonstrations that make expert analytical processes visible to students (Attas, 2018). By externalizing the

reasoning behind theoretical analysis, instructors can scaffold complex cognitive processes and gradually build students' analytical proficiency. In addition, the flipped model promotes autonomy in how students engage with preparatory materials, strengthens competence through guided application, and fosters relatedness through collaborative in-class activities. It also reflects UDL principles by allowing flexibility in how information is accessed and how understanding is demonstrated.

Individually, each of these pedagogical strategies addresses a specific instructional challenge in music theory education. Collectively, however, they form an integrated pedagogical system: SBG clarifies learning goals, JiTT provides diagnostic insight into student understanding, and the flipped classroom creates the conditions for active engagement with theoretical concepts. Their combined effectiveness can be interpreted through SDT, which highlights the roles of autonomy, competence, and relatedness in sustaining motivation, as well as through UDL, which emphasizes flexible and inclusive learning environments capable of supporting diverse learners.

Despite these advantages, the adoption of these strategies presents practical challenges. Teachers must carefully balance in-class and out-of-class workloads, ensure the pedagogical quality of asynchronous learning materials, and guide students through transitions toward unfamiliar assessment structures. Consequently, institutional support—particularly in the form of professional development, curriculum coordination, and technological infrastructure—remains essential for sustainable implementation.

Overall, the convergence of SBG, JiTT, and flipped learning illustrates a broader pedagogical transformation within music theory education. By linking transparent learning goals, adaptive feedback mechanisms, and collaborative analytical practice, these strategies demonstrate how music theory instruction can evolve beyond abstract exposition toward a more participatory and learner-responsive model of higher education.

RQ3: What Challenges Do Students Encounter, and What Solutions Are Implemented in Music Theory Education?

The two studies focused on the challenges and strategies to address music theory education. A review and details of the two studies are shown in Table 8, including the year of publication, thematic focus, research objectives, geographic context, and details of methodological data for each article.

Challenges in music theory education arise from a dynamic interplay between external structures and internal learner needs. As reported in Slovenia, rigid teaching

Table 8. Challenges and Solution Strategies in Music Theory Education.

Researcher/ Date	Country of origin	Methodology	Participants	Findings	Implications
Kavčič Pucihar et al. (2024)	Slovenia	Qualitative content analysis (semi- structured focus groups)	Students (6), parents (6), instrumental music teachers (6), music theory teachers (6), primary music school principals (5)	External factors: teaching methods, social environment, music school curriculum, resource constraints, overall workload Internal factors: autonomy, competence, relatedness, individual differences	The study highlights the importance of a supportive environment to meet individual needs and interests, which is critical to improving the overall experience of music education and reducing dropout rates.
Ebbini (2023)	Jordan	Descriptive research methodology (author's own teaching experience as primary resource, combined with literature)	Music theory teachers and students	Interactive, pedagogical, and cognitive-based difficulties encountered in teaching online; challenges of teaching music theory; lack of technological features in existing software and platforms needed to effectively teach music theory	The study suggests a number of pedagogical strategies to assist online music theory teachers and emphasizes the urgent need for institutional and technological interventions to ensure that future online learning efforts are effective and successful.
Gutierrez (2018)	USA	Mixed Methods Study	291 graduates of music programs recruited via Reddit	Students highly value theory knowledge but report major shortcomings in integration, diversity, and creativity in their theory classes	Curriculum reform should prioritize student feedback, integrating diverse music and creative activities

methods, overloaded curricula, limited resources, and the pressures of the broader social environment interact with individual factors such as autonomy, competence, and relatedness (Kavčič Pucihar et al., 2024). These structural and motivational issues are not isolated; indeed, large-scale survey research with music graduates in the United States has corroborated that a lack of integration with students' goals, insufficient diversity in musical content, and a deficit of creative engagement are primary sources of frustration (Gutierrez, 2018). Because these conditions often fail to align, students' motivation weakens, and dropout risks increase. Therefore, the challenge is not merely about teaching content but about creating an environment that supports individual differences.

However, challenges are not limited to traditional classroom contexts. In Jordan, online music theory instruction revealed further difficulties: teachers struggled to adapt abstract concepts into digital forms, and existing platforms lacked the necessary tools to foster interactivity (Ebbini, 2023). Because online teaching depends heavily on technological mediation, its shortcomings directly translate into pedagogical barriers. Consequently, teachers often rely on improvisation, which limits systematic improvement. Yet, as Uludag and Satir (2025) argue, effective digital transformation in

education requires institutional support and targeted investment; individual teacher effort alone is insufficient.

Taken together, these studies highlight a paradox. On one hand, music theory education aspires to cultivate autonomy, creativity, and expressive interpretation. On the other, both structural overload in traditional settings and technological inadequacies in online environments constrain these very outcomes. The implication is clear: solutions must be multidimensional. Reflective teaching practices can help teachers adapt, but without institutional reform to balance workloads and without technological platforms tailored for music pedagogy, progress will remain fragmented.

Therefore, three strategic pathways emerge. First, hybrid teaching models should be designed to combine the stability of face-to-face methods with the flexibility of digital delivery, thereby enhancing competence and autonomy. Second, curricula should be restructured in line with UDL principles so that assessments and learning tasks are accessible to diverse learners rather than privileging a narrow range of abilities. Third, institutions must commit to sustained professional development and investment in purpose-built technologies, equipping teachers with both the strategies and the infrastructure necessary to succeed. Only by addressing these

Table 9. Assessment and Effectiveness of Music Theory Education.

Researcher/ Date	Country of origin	Methodology	Participants	Findings	Implications
Barić and Begić (2022)	Croatia	Quantitative study (questionnaire)	101 teachers of theory and musicology	Teachers highly value their educational competence; the need for personal and professional development; importance of maintaining quality relationships with students increases with work experience.	Teachers' educational competence is related to work experience; and the need for continuous professional development.
Buonviri and Paney (2015)	USA	Online survey research	875 U.S. AP music theory teachers (398 completed survey)	Teachers' preference for a musical system that emphasizes scale degree functions and rhythmic systems; AP exams have an impact on their teaching; and more instructional time and better preparation in listening skills are needed.	High school music theory teachers use a variety of approaches to teaching melodic dictation, requiring additional instructional resources and time.
Nedelcuț and Ghișa (2020)	Romania	Quantitative study (student feedback questionnaire)	No specific number of participants provided, students of the National Conservatory "Gheorghe Dima"	Students' evaluation of the distance education system, including tutors, course coordinators and the way the distance education system is organized.	The distance education system needs to be adapted to the needs of the students in order to improve the effectiveness of the teaching.
Nolte (2019)	USA	Quantitative study	Two cohorts of undergraduate music majors (pre- and post-curricular reform)	Non-keyboard majors face challenges of insufficient foundational skills, including unfamiliarity with treble and bass clefs, key signatures, and scales, resulting in slow progress in theory courses.	Through the mandatory piano preparatory course—a structural reform—students' foundational skill challenges were systematically addressed with remarkable results.

interconnected challenges at instructional, institutional, and technological levels can music theory education evolve from being perceived as rigid and abstract to becoming an engaging, inclusive, and sustainable foundation for lifelong musicianship.

RQ4: How is Student Learning Assessed, and How Effective is Music Theory Education?

Three studies focused on the assessment and effectiveness of music theory education. A review and details of the three studies are shown in Table 9, including the year of publication, thematic focus, study objectives, geographic context, and details of methodological data for each article.

Assessment occupies a central role in music theory education, functioning not only as a measure of achievement but also as a feedback system that informs pedagogy and curriculum. When designed effectively, it connects learning objectives with outcomes, supports instructional adaptation, and ensures that abstract concepts are translated into meaningful practice. The

literature highlights three key levels at which assessment shapes effectiveness: teacher self-reflection, student feedback, and external benchmarks (Nolte, 2019).

Teacher self-assessment, identified as a catalyst for professional growth, reveals both strengths and challenges. Croatian instructors rated their competence positively but acknowledged difficulties in building relationships and adapting to individual needs; with experience, many shifted toward more student-centered methods, suggesting that reflective practice enhances responsiveness (Barić & Begić, 2022). From the perspective of Self-Determination Theory (SDT), teacher reflection strengthens relatedness by prioritizing student connections, while from a Universal Design for Learning (UDL) standpoint, it supports flexibility by prompting instructors to diversify strategies for different learners.

Student evaluations add another essential dimension. Research from the "Gheorghe Dima" National Music Academy showed that learners valued blended approaches but pointed to gaps in resources and interaction (Nedelcuț & Ghișa, 2020). Such feedback provides instructors with actionable insights and increases

transparency in the learning process. When students perceive assessment as fair and supportive, their motivation and persistence improve—an outcome that can again be explained through SDT (by reinforcing autonomy and competence) and UDL (by aligning evaluation with diverse learner needs).

At the level of external benchmarks, studies of Advanced Placement (AP) Music Theory highlight how standardized frameworks shape classroom practice. Teachers reported using strategies such as scale-degree recognition and rhythm training to align with AP exam requirements, but they also noted constraints in time and resources that limited the scope of aural skills assessment (Buonviri & Paney, 2015). This illustrates a tension between preparing students for external evaluations and fostering broader outcomes like creativity or critical thinking. Here, UDL reminds us that overreliance on narrow measures reduces opportunities for varied demonstration of learning, while SDT suggests that externally imposed goals may undermine intrinsic motivation if not balanced with autonomy-supportive practices.

Taken together, these studies show that assessment in music theory must balance summative and formative purposes. Self-reflection enhances instructional quality, student feedback strengthens engagement, and standardized assessments provide accountability, but effectiveness depends on integrating these functions coherently. Narrow measures of recall or technical proficiency risk overlooking higher-order outcomes such as creativity, aural sensitivity, and expressive interpretation, which are central to professional musicianship.

For assessment to be truly effective, it must be multi-dimensional. Formative strategies that provide continuous feedback should complement summative evaluations, ensuring that both technical mastery and artistic expression are recognized. Teacher self-assessment and student feedback need to be embedded in iterative cycles of improvement, while external exams should be balanced with opportunities for authentic, autonomy-supportive learning. Viewed through the lenses of UDL and SDT, such an approach highlights assessment not merely as measurement, but as a dynamic mechanism that cultivates inclusivity, supports motivation, and sustains meaningful engagement in music theory education.

Discussion

Evolving Trends in Undergraduate Music Theory Education

This review synthesizes empirical studies examining learning needs and pedagogical practices associated with undergraduate music theory education, highlighting the interconnections among instructional content, teaching

strategies, learning environments, and assessment practices. The findings suggest that music theory instruction is currently undergoing a pedagogical transition, moving beyond traditional lecture-centered approaches toward models that emphasize technological integration, learner-centered pedagogy, and flexible instructional design. These developments reflect broader trends in higher education that prioritize active engagement, individualized learning pathways, and adaptive teaching strategies. In this context, the emerging instructional practices identified in the reviewed studies align closely with the principles of Self-Determination Theory (SDT) and Universal Design for Learning (UDL), both of which emphasize the importance of autonomy, competence, and inclusive learning environments in fostering meaningful educational experiences (CAST, 2018; Deci & Ryan, 2000).

It should also be noted that while the primary focus of this review is undergraduate music theory education, several included studies examined broader music-learning contexts, such as online music communities or school-level teaching practices. These studies were retained because they reveal pedagogical processes related to how music theory knowledge is learned, applied, and transmitted across different learning environments. As such, their findings provide transferable insights that can inform instructional design and pedagogical innovation in undergraduate music theory courses.

Shift From Abstract Knowledge to Contextualized Learning

The reviewed studies indicate a significant shift from the traditional, abstract transmission of knowledge toward more applied, contextualized learning. Embedding theory within creative practices—such as composition, improvisation, and analysis of popular music—allows students to recognize the relevance of theoretical concepts to their artistic goals (Duker et al., 2015; Li, 2020). This transition aligns with constructivist perspectives, which argue that learning is most effective when students can connect theoretical knowledge to real-world tasks (Bruner, 1996). Pedagogical approaches such as flipped classrooms, standards-based grading, and just-in-time teaching promote active student engagement, timely feedback, and self-directed learning. These strategies directly contribute to the development of competence and autonomy—core tenets of SDT (Ryan & Deci, 2000).

Challenges and Opportunities in Learning Environments

Learning environments have also evolved to support these new teaching methodologies. The integration of digital tools, including mobile applications and auto-

accompaniment software, facilitates personalized learning and provides students with immediate feedback, enhancing both cognitive and emotional engagement. However, challenges persist in ensuring equitable access to these resources and in providing adequate teacher preparation for effectively incorporating technology into the curriculum (Buonviri & Paney, 2020; Uludag & Satir, 2025). These findings underscore the necessity for institutions to invest in both infrastructure and training to ensure that both students and educators can fully leverage digital tools, creating a more inclusive and dynamic learning environment.

Assessment Practices and Future Directions

Assessment practices, traditionally focused on recall and technical proficiency, are gradually expanding to capture creativity, critical thinking, and musical expression. However, external assessments such as standardized exams continue to influence curricular priorities, sometimes limiting the scope of what is taught and how it is assessed (Buonviri & Paney, 2015). The challenge moving forward is to develop assessment frameworks that are both rigorous and reflective of the diverse skills and competencies required in music theory, including creativity and critical engagement with the material (Nedelcuț & Ghișa, 2020). This shift is essential for ensuring that assessments measure more than just technical knowledge but also creative and analytical abilities.

In conclusion, this review calls for systemic coherence in music theory instruction, where content, pedagogy, technology, and assessment are aligned to support the diverse needs of learners. The theoretical frameworks of SDT and UDL provide a valuable lens through which to understand these changes, highlighting the importance of autonomy, competence, and inclusive access in shaping music theory education. The review advocates for continued research into how these frameworks can be integrated into music theory curricula, particularly in diverse educational contexts, and how they can guide future pedagogical practices and policy development.

Limitations

Despite the methodological rigor adhered to in this review, certain limitations should be acknowledged. First, the majority of the included studies were conducted within Western higher education contexts. Second, to ensure methodological consistency and reliability, this review deliberately focused on research published in peer-reviewed English-language journals. While this inclusion criterion strengthens analytical coherence and credibility, it may have inadvertently excluded valuable insights from gray literature—such as dissertations,

institutional reports, and non-English publications. This relative homogeneity in data sources may, to some extent, limit the diversity of viewpoints and the comprehensiveness of the conclusions.

Conclusion

This systematic review synthesizes 15 studies published between 2015 and 2024, offering a comprehensive analysis of the evolving landscape of undergraduate music theory education. The findings highlight a shift from traditional, abstract teaching methods to more learner-centered, flexible pedagogies, such as flipped classrooms and just-in-time teaching, which better address the diverse needs of students. The integration of technology, including mobile apps and auto-accompaniment software, has proven effective in fostering engagement and supporting personalized learning. However, challenges remain, such as unequal access to digital tools and the need for improved teacher preparation. The review underscores the importance of aligning instructional content, pedagogy, and assessment to create inclusive, adaptive learning environments. Future research should focus on longitudinal studies and explore the impact of hybrid instructional models, incorporating student feedback and affective outcomes. Ultimately, the review calls for music theory pedagogy to evolve from content transmission to adaptive, evidence-based practices that foster both conceptual mastery and creative engagement.

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Ethical Considerations

Since this study does not involve human participants or animal experiments, ethics approval is not required.

Consent to Participate

As this study does not involve human participants, consent to participate is not applicable.

Consent for Publication

This study does not involve personal data or privacy concerns, so consent to publish is not required.

Author Contributions

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This study is a systematic literature review, and all data used were sourced from publicly available academic literature.

Supplemental Material

Supplemental material for this article is available online.

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